

Python Assignments and Exercises Report

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Exercises 1-41: Fundamentals, Files, OOP, Data, Flask, and MQTT

Repository: <https://github.com/chittarad06-cell/ECRIO-Exercises>

Abstract

This report summarizes the completed Python assignment exercises. The exercises are grouped by topic so the repository is easy to review and the learning progression is clear.

Exercise s	File or folder	Implementation strength
1-4	01_python_basics.py	functions, recursion, lambda, lists, tuples, range, break, continue
5-6	02_files_regex_encoding.py	regex, pathlib, Base64, ASCII, URL encoding, ROT13, CSV, JSON, binary files
7-12	03_oop_logging_collections.py	dataclass, inheritance, logging, deque, custom exceptions
13-21	04_data_analysis.py	NumPy, pandas, filtering, groupby, pivot table, normalization, rolling average
22-30	flask_portfolio_app	routes, templates, forms, sessions, JSON POST validation, Docker/Procfile
31-41	mqtt_examples	MQTT report, topic design, publisher/subscriber, callbacks, QoS, threading

Exercises 1-4: Python Fundamentals

- variables and type conversion
- functions, recursion, lambda
- range-based repetition and filtering

These exercises establish the foundation: how to store values, transform values, reuse logic, and control program flow.

Exercises 5-6: Files, Regex, and Encoding

- regex for pattern extraction
- pathlib for file handling
- Base64/ASCII/URL/ROT13 round trips
- CSV, JSON, and binary file examples

This section shows Python as an automation tool for real text and file processing tasks.

Exercises 7-12: OOP and Reliability

- inheritance and method overriding
- encapsulation and class methods
- logging levels
- stack, queue, deque, custom exceptions

This section connects Python syntax with software quality through classes, logging, collections, and safe error handling.

Exercises 13-21: Data Analysis

- NumPy statistics
- pandas cleaning and grouping
- pivot table and normalization
- time-series rolling average

This section shows why data must be cleaned, transformed, summarized, and checked before conclusions are drawn.

Exercises 22-30: Flask

- routes, query strings, templates
- forms and sessions
- JSON POST validation
- Docker and deployment files

This section demonstrates how Python can serve web pages, accept user input, validate API data, and prepare for deployment.

Exercises 31-41: MQTT

- MQTT vs HTTP comparison
- smart-home topic hierarchy
- publisher/subscriber callbacks
- QoS, security, and threading notes

This section introduces event-driven IoT communication and shows why MQTT fits repeated sensor messages better than normal request-response HTTP.

Conclusion

The exercises show a complete path from basic Python to practical automation, data analysis, web applications, and IoT communication. The work emphasizes choosing the correct construct for each task.

Submission Evidence

- Repository includes code for exercises 1-41 from the visible assignment sheet.
- README explains topic grouping and why specific constructs were used.
- This report documents implementation choices and learning outcomes.
- VIDEO_GUIDE.md is included to support the short explanation video.